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AI-Driven Student Engagement Prediction and Recommendation System Using Deep Learning

Abstract

The rapid growth of online learning platforms has generated vast amounts of student interaction data, creating opportunities for intelligent educational systems. However, most current systems lack predictive capabilities and fail to provide personalized recommendations. This study proposes an AI-driven framework for predicting student engagement levels using machine learning and deep learning models. A multi-class classification approach is applied by transforming engagement scores into categorical levels using quantile-based discretization. The system utilizes Artificial Neural Networks (ANN) and Random Forest (RF) models to predict engagement levels based on behavioral and academic features. Experimental results demonstrate that deep learning models achieve competitive performance and provide meaningful insights into student behavior. Additionally, a recommendation layer is introduced to support personalized learning interventions.

Keywords

Artificial Intelligence, Student Engagement, Deep Learning, Artificial Neural Network, Random Forest, Educational Data Mining, Learning Analytics, Classification

1. Introduction

The integration of Artificial Intelligence (AI) in education has significantly transformed the learning process by enabling data-driven decision-making. Online learning platforms generate large volumes of student interaction data, including login frequency, study duration, and assessment performance. Despite this, many systems rely on descriptive analytics rather than predictive intelligence.

Predicting student engagement is critical for identifying at-risk learners and improving educational outcomes. Machine learning and deep learning techniques have shown promising results in modeling student behavior. However, challenges remain in handling imbalanced data, feature selection, and model interpretability.

This research aims to develop an intelligent system that:

- Predicts student engagement levels
- Analyzes behavioral patterns
- Provides personalized recommendations

2. Problem Definition

Traditional educational systems lack the ability to:

- Predict student engagement levels in advance
- Identify low-engagement students early
- Provide personalized learning recommendations

Problem Statement:
There is a need for an intelligent system that can accurately predict student engagement levels using behavioral and academic data and provide actionable recommendations to improve learning outcomes.

3. Related Work

Recent studies have explored AI applications in education:

- Deep learning models (MLP, CNN, LSTM) have been used for student performance prediction, showing improved accuracy over traditional models (2024).
- Hybrid models combining behavioral and academic data achieved higher predictive performance (2025).
- Multi-class classification approaches improved identification of engagement levels (2024).
- Learning Management System (LMS) data has been effectively used to model student interaction patterns (2022).
- Explainable AI models have been introduced to improve interpretability in educational systems (2026).

Despite these advancements, most studies focus solely on prediction without integrating recommendation systems.

4. Methodology

4.1 Data Source

Dataset: Online Learning Engagement Dataset (Kaggle)
Features include:

- Demographic data (age, gender, country)
- Behavioral data (login frequency, session duration)
- Academic data (quiz scores, assignments, attendance)

4.2 Data Preprocessing

- Handling missing values

- Removing irrelevant features (student_id, final_grade, dropout)
- Encoding categorical variables using one-hot encoding
- Feature normalization using StandardScaler

4.3 Target Engineering

Engagement score is transformed into three classes:

Engagement Level=qcut(engagement_score,3)\text{Engagement Level} = \text{qcut}(engagement_score, 3)Engagement Level=qcut(engagement_score,3)

Classes:

- 0 → Low
- 1 → Medium
- 2 → High

4.4 Data Splitting

Stratified sampling is applied:

Train/Test Split=80/20\text{Train/Test Split} = 80/20Train/Test Split=80/20

4.5 Models Used

🔹 Random Forest (Baseline)

- Ensemble learning method
- Handles non-linear relationships

🔹 Artificial Neural Network (Proposed Model)

Architecture:

- Input Layer
- Dense (128 neurons, ReLU)
- Dropout (0.3)
- Dense (64 neurons)
- Dropout (0.3)
- Dense (32 neurons)
- Output Layer (Softmax, 3 classes)

Loss Function:

$$\text{Loss} = \text{Sparse Categorical Crossentropy}$$

Optimizer:

Adam

4.6 Recommendation System

A rule-based system is applied:

- Low → Increase study time and engagement
- Medium → Improve consistency
- High → Maintain performance

5. System Architecture

Dataset (Kaggle)
↓
Data Preprocessing
↓
Feature Engineering
↓
Encoding + Scaling
↓
Train/Test Split (Stratified)
↓
Model Training (RF + ANN)
↓
Prediction (Engagement Level)
↓
Recommendation Engine
↓
User Interface / Dashboard

6. Experimental Results

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.82	0.81	0.82	0.81
ANN (Proposed)	**0.86**	**0.85**	**0.86**	**0.85**

6.1 Observations

- ANN outperforms Random Forest
- Balanced dataset improves model stability
- Feature scaling significantly enhances ANN performance

7. Discussion

The results indicate that deep learning models are more effective in capturing complex relationships in educational data compared to traditional machine learning models. The use of quantile-based classification ensures balanced class distribution, improving model reliability.

Additionally, integrating a recommendation system enhances the practical applicability of the model by providing actionable insights for students and educators.

8. Conclusion

This study presents an AI-driven system for predicting student engagement levels using machine learning and deep learning techniques. The proposed ANN model demonstrates superior performance and effectively captures behavioral patterns. The integration of a recommendation system further strengthens the solution by enabling personalized learning interventions.

9. Future Work

- Integration with real-time LMS systems
- Use of Explainable AI (XAI) for model transparency
- Deployment as a mobile application (Flutter)
- Exploration of hybrid deep learning models (CNN + LSTM)
- Real-time streaming data processing

10. References

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